STA3000F: Advanced Theory of Statistics **Fall 2026**

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1 Covering and packing numbers

Throughout this section, covering balls are open, and an ϵ-packing has pairwise distances at least ϵ.

Lemma 6.1 (duality)**.** *For any nonempty totally bounded metric space (K, ρ) and $\epsilon > 0$, we have*

$$M(K; \rho, 2\epsilon) \overset{(i)}{\leq} N(K; \rho, \epsilon) \overset{(ii)}{\leq} M(K; \rho, \epsilon).$$

Roughly speaking, if you do not care about constant factors in the radius, covering number and packing number are equivalent.

Proof of (i): Let $(\theta_i)_{i=1}^M$ be a 2ϵ-packing of K, and let $(\psi_i)_{i=1}^N$ be an ϵ-covering of K. We need to show that $M \leq N$. By 2ε-packing, we know that $\mathbb{B}_\rho(\theta_i, \epsilon) \cap \mathbb{B}_\rho(\theta_j, \epsilon) = \emptyset$ for $i \neq j$. On the other hand, since $(\psi_i)_{i=1}^N$ is an ϵ-covering, for each θ_i, there exists some ψ_j such that $\theta_i \in \mathbb{B}_\rho(\psi_j, \epsilon)$. For different θ_i and θ_ℓ, they cannot be covered by the same ψ_j, otherwise we would have $\rho(\theta_i, \theta_\ell) \leq \rho(\theta_i, \psi_j) + \rho(\psi_j, \theta_\ell) < 2\epsilon$, which contradicts the 2ϵ-packing property. So for each i, there exists j such that $\theta_i \in \mathbb{B}_\rho(\psi_j, \epsilon)$, and different i's correspond to different j's. This shows that $M \leq N$.

Proof of (ii)**:** Let $(\theta_i)_{i=1}^M$ be a maximal ε-packing. That means we cannot add any more points to it and maintain the ε-packing property. Therefore, for any $\theta \in K$, there exists some θ_i such that $\rho(\theta, \theta_i) < \varepsilon$, otherwise we can add θ to the packing. This shows that $(\theta_i)_{i=1}^M$ is also an ε-covering of K. Therefore, $N(K; \rho, \epsilon) \leq M(K; \rho, \epsilon)$.

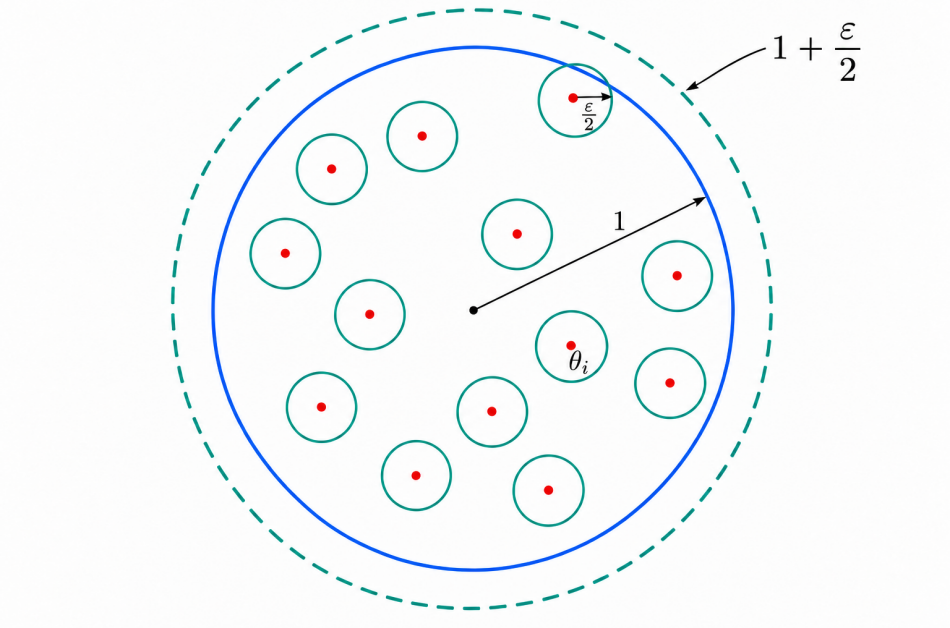
1.1 Examples

Let $K = \mathbb{B}_2^d(1)$, the unit ball in $\mathbb{R}^d$ under ℓ_2 norm. We want to compute the covering number $N(K; \|\cdot\|_2, \epsilon)$.

Upper bound: Suppose $(\theta_i)_{i=1}^M$ is an ϵ-packing of K. Then we have

$$\bigcup_{i=1}^M \mathbb{B}_2^d(\theta_i, \epsilon/2) \subseteq \mathbb{B}_2^d(1 + \epsilon/2).$$

The geometry behind this inclusion is illustrated below: the small balls are disjoint, and each is contained in the enlarged ball.



Volume comparison for an ϵ-packing. The blue circle is the unit ball; the dashed circle has radius $1 + \epsilon/2$. Each small ball has radius $\epsilon/2$ and is centered at a packing point. Schematic, not to scale.

This implies that

$$M \cdot \mathrm{vol}(\mathbb{B}_2^d(\epsilon/2)) \leq \mathrm{vol}(\mathbb{B}_2^d(1 + \epsilon/2)),$$

which gives us

$$M \leq \left(1 + \frac{2}{\epsilon}\right)^d,$$

and by duality, we have

$$N(K; \|\cdot\|_2, \epsilon) \leq \left(1 + \frac{2}{\epsilon}\right)^d.$$

Lower bound: Suppose $(\psi_i)_{i=1}^N$ is an ϵ-covering of K. Then we have

$$\mathrm{vol}(\mathbb{B}_2^d(1)) \leq N \cdot \mathrm{vol}(\mathbb{B}_2^d(\epsilon)),$$

which gives us

$$N \geq \frac{\mathrm{vol}(\mathbb{B}_2^d(1))}{\mathrm{vol}(\mathbb{B}_2^d(\epsilon))} = \left(\frac{1}{\epsilon}\right)^d.$$

The ratio between these displayed upper and lower bounds is $(2+\epsilon)^d$. In statistical applications, we usually use the log covering number. Uniformly for $0 < \epsilon \leq 1/2$, the two bounds imply

$$\log N(K; \|\cdot\|_2, \epsilon) \asymp d \log(1/\epsilon).$$

2 Application to M-estimation

Recall that we are interested in

$$\widehat{\theta}_n \in \arg\min_{\theta\in\Theta} F_n(\theta), \quad \theta^* \in \arg\min_{\theta\in\Theta} F(\theta).$$

We need to show uniform convergence of $F_n(\theta)$ to $F(\theta)$.

Theorem 6.1 (Wald)**.** *Let $X_1, \ldots, X_n$ be i.i.d., let (Θ, ρ) be a nonempty compact metric space, and let $f(x;\theta)$ be measurable in x and continuous in θ for each x. Write $F_n(\theta) = n^{-1}\sum_{i=1}^n f(X_i;\theta)$ and $F(\theta) = \mathbb{E}[f(X;\theta)]$. Suppose furthermore that*

$$\mathbb{E}\Big[\sup_{\theta\in\Theta} |f(X;\theta)|\Big] < +\infty.$$

(i.e., the class has an integrable envelope function). Then we have

$$\sup_{\theta\in\Theta} |F_n(\theta) - F(\theta)| \xrightarrow{p} 0.$$

Proof. Let $G(X) = \sup_{\theta\in\Theta} |f(X;\theta)|$, and define the uniform modulus

$$M_\delta(X) = \sup_{\substack{\theta,\theta'\in\Theta \\ \rho(\theta,\theta')\le\delta}} |f(X;\theta) - f(X;\theta')|.$$

For each X, continuity on the compact set Θ implies uniform continuity, so $M_\delta(X) \to 0$ as $\delta \downarrow 0$. Also $M_\delta(X) \le 2G(X)$. Dominated convergence therefore gives $\mathbb{E}[M_\delta(X)] \to 0$. In particular, F is continuous.

For fixed $\delta > 0$, choose a finite δ-cover $\{\theta_1, \ldots, \theta_N\}$ of Θ. The triangle inequality gives

$$\sup_{\theta\in\Theta} |F_n(\theta) - F(\theta)| \le \underbrace{\max_{1\le i\le N} |F_n(\theta_i) - F(\theta_i)|}_{T_1} + \mathbb{E}[M_\delta(X)] + \underbrace{\frac{1}{n}\sum_{j=1}^n M_\delta(X_j)}_{T_3}.$$

For this fixed finite cover, $T_1 \to 0$ in probability by the law of large numbers and a finite union bound. The same law gives $T_3 \to \mathbb{E}[M_\delta(X)]$ in probability. Given any $\varepsilon > 0$, first choose δ so that $2\mathbb{E}[M_\delta(X)] < \varepsilon/2$, and then let $n \to \infty$. The displayed bound proves the claimed uniform convergence.

Remarks:

1. This only needs very weak conditions, but only proves asymptotic convergence.
2. Using this result, and in combination with basic inequality, we have $F(\widehat{\theta}_n) - F(\theta^*) \xrightarrow{p} 0$. Furthermore, under the same setup, F is continuous, and assuming that θ^* is unique minimizer, then we have $\widehat{\theta}_n \xrightarrow{p} \theta^*$.
3. Can we turn this into a more quantitative bound?

For the term T_1, we note that

$$\begin{aligned}
&\mathbb{P}\Big(\max_{i=1,\cdots,N} |F_n(\theta_i) - F(\theta_i)| > \varepsilon\Big) \\
&\quad\le \sum_{i=1}^{N} \mathbb{P}(|F_n(\theta_i) - F(\theta_i)| > \varepsilon) \\
&\quad\le N \cdot \max_{i=1,\cdots,N} \mathbb{P}(|F_n(\theta_i) - F(\theta_i)| > \varepsilon).
\end{aligned}$$

If $0 \le f(x;\theta) \le 1$, then we can use Hoeffding's inequality to get

$$\mathbb{P}\Big(\max_{i=1,\cdots,N} |F_n(\theta_i) - F(\theta_i)| > \varepsilon\Big) \le 2Ne^{-2n\varepsilon^2}.$$

By taking $\varepsilon = \sqrt{\frac{\log(2N/\delta)}{2n}}$, we have the high-probability bound.

On the other hand, if f is unbounded (or even heavy-tailed), we cannot directly apply Hoeffding's inequality. If the variances are uniformly bounded, a naive Chebyshev union bound gives ε of order $\sqrt{N/n}$ at a fixed failure probability, which is not useful when N is large.

Symmetrization and Rademacher complexity give a more flexible approach; quantitative bounds still require suitable moment or envelope conditions.

For the term T_3, we can bound it as long as we have quantitative bound on the modulus-of-continuity of $f(X;\theta)$ in θ. But on the other hand, this type of one-step discretization argument is fundamentally sub-optimal. We can do better by using "chaining method", which is a multi-step discretization argument.

3 Symmetrization and chaining methods

3.1 Symmetrization method

Introducing the notations:

$$P_n f = \frac{1}{n}\sum_{i=1}^{n} f(X_i), \quad Pf = \mathbb{E}[f(X)], \quad (P_n - P)f = P_n f - Pf.$$

For notation simplicity, we drop the parametrization by θ and write $f(X)$ instead of $f(X;\theta)$. We are interested in bounding

$$\mathbb{E}\Big[\sup_{f\in\mathcal{F}} |(P_n - P)f|\Big].$$

Here $\mathcal{F} = \{x \mapsto f(x;\theta) : \theta \in \Theta\}$ is the function class induced by the parametrization. Assume this class is nonempty, has a measurable integrable envelope, and is pointwise separable: a countable subclass approximates every f pointwise. These conditions ensure that the suprema and expectations below are measurable. The Rademacher variables are independent of the data.

The "symmetrized version" of the above quantity is

$$\mathcal{R}_n(\mathcal{F}) := \mathbb{E}\Big[\sup_{f\in\mathcal{F}} \Big|\frac{1}{n}\sum_{i=1}^{n} \varepsilon_i f(X_i)\Big|\Big], \quad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \text{Rademacher}(\pm 1),$$

which is called the Rademacher complexity of the function class $\mathcal{F}$. The Rademacher complexity is a measure of the richness of the function class $\mathcal{F}$, and it is often easier to bound than the original quantity, because we can bound it by conditioning on $X_1, \cdots, X_n$ and using concentration inequalities for Rademacher random variables.

Theorem 6.2 (symmetrization)**.** *We have*

$$\mathbb{E}\Big[\sup_{f\in\mathcal{F}} |(P_n - P)f|\Big] \le 2\mathcal{R}_n(\mathcal{F}).$$

Proof: Let $X_1', \cdots, X_n' \overset{\text{i.i.d.}}{\sim} \mathbb{P}$ be an independent copy of $X_1, \cdots, X_n$. Then we have

$$\begin{aligned}\mathbb{E}\Big[\sup_{f\in\mathcal{F}} |(P_n - P)f|\Big] &= \mathbb{E}\Big[\sup_{f\in\mathcal{F}} \Big|\frac{1}{n}\sum_{i=1}^n \big(f(X_i) - \mathbb{E}[f(X_i')]\big)\Big|\Big] \\ &\overset{\text{(Jensen)}}{\le} \mathbb{E}\Big[\sup_{f\in\mathcal{F}} \Big|\frac{1}{n}\sum_{i=1}^n \big(f(X_i) - f(X_i')\big)\Big|\Big]\end{aligned}$$

We know that by introducing Rademacher random variables $\varepsilon_i \overset{\text{i.i.d.}}{\sim}$ Rademacher(± 1), we have

$$\big(f(X_i) - f(X_i')\big)_{i=1}^n \overset{d}{=} \big(\varepsilon_i(f(X_i) - f(X_i'))\big)_{i=1}^n,$$

jointly over $f \in \mathcal{F}$: each pair (X_i, X_i') can be swapped independently without changing the joint law. Thus

$$\text{above} = \mathbb{E}\Big[\sup_{f\in\mathcal{F}} \Big|\frac{1}{n}\sum_{i=1}^n \varepsilon_i(f(X_i) - f(X_i'))\Big|\Big] \le 2\mathcal{R}_n(\mathcal{F}).$$

Application of symmetrization method: We can reason conditionally on $X_1, \cdots, X_n$, and Rademacher random variables are bounded and nicely behaved in terms of concentration inequalities.

In particular, for any $A \subseteq \mathbb{R}^n$, we define the Rademacher complexity of A as

$$R_n(A) := \mathbb{E}\Big[\sup_{a\in A} \Big|\frac{1}{n}\sum_{i=1}^n \varepsilon_i a_i\Big|\Big], \quad \varepsilon_i \overset{\text{i.i.d.}}{\sim} \text{Rademacher}(\pm 1).$$

This has nothing to do with the data or statistical problem, and is purely a combinatorial measure of the complexity of the set A.

Following the arguments in the previous section, we can bound it using concentration inequality.

For finite A: for any $\lambda > 0$, by Jensen's inequality,

$$\begin{aligned}R_n(A) &\le \frac{1}{\lambda}\log \mathbb{E}\Big[\max_{a\in A\cup(-A)} \exp\Big(\lambda\frac{1}{n}\sum_{i=1}^n \varepsilon_i a_i\Big)\Big] \\ &\le \frac{1}{\lambda}\log\Big\{\sum_{a\in A\cup(-A)} \mathbb{E}\Big[\exp\Big(\lambda\frac{1}{n}\sum_{i=1}^n \varepsilon_i a_i\Big)\Big]\Big\} \\ &\le \frac{\log(2|A|)}{\lambda} + \frac{\lambda}{2n^2}\max_{a\in A}\sum_{i=1}^n a_i^2 \\ &\overset{\text{(choosing optimal } \lambda)}{=} \sqrt{\frac{2\log(2|A|)}{n}} \cdot \max_{a\in A}\sqrt{\frac{\sum_{i=1}^n a_i^2}{n}}.\end{aligned}$$

3.2 Discretization and chaining

Following the same idea as before, we bound $R_n(A)$ for a nonempty bounded set $A \subseteq \mathbb{R}^n$. Write

$$\|a\|_n = \sqrt{\frac{1}{n}\sum_{i=1}^n a_i^2}.$$

For all diameter-based bounds below, assume $0 \in A$. This anchor is necessary for an absolute supremum: a nonzero singleton has diameter zero but positive Rademacher complexity. For a general bounded set and any $a_0 \in A$, translation instead gives

$$R_n(A) \le R_n(A - a_0) + \frac{\|a_0\|_n}{\sqrt{n}}.$$

One-step discretization: Let $(a^{(i)})_{i=1}^N$ be a minimum-cardinality ε-covering of A with respect to this normalized metric, and write $N(\varepsilon) = N(A; \|\cdot\|_n, \varepsilon)$. Denote $\pi(a) := \arg\min_{a^{(i)}} \left\|a - a^{(i)}\right\|_n$ as the projection of a onto the covering set. Then we have

$$\begin{aligned} R_n(A) &\le \underbrace{\mathbb{E}\Big[\sup_{a\in A} n^{-1}|\varepsilon^\top (a - \pi(a))|\Big]}_{\text{(C.–S.)} \quad \le \sup_{a\in A}\|a-\pi(a)\|_n} + \underbrace{\mathbb{E}\Big[\sup_{a\in A} n^{-1}|\varepsilon^\top \pi(a)|\Big]}_{\le \sqrt{\frac{2\log(2N)}{n}}\cdot \mathrm{diam}(A)} \\ &\le \varepsilon + \sqrt{\frac{2\log(2N(\varepsilon))}{n}} \cdot \mathrm{diam}(A). \end{aligned}$$

Is this argument tight?

Consider A being the unit ball under $\|\cdot\|_n$ in a d-dimensional subspace of $\mathbb{R}^n$, where $1 \le d \le n$ (as in a d-dimensional linear model for $\mathcal{F}$). By choosing $\varepsilon = O(1/n)$, we have

$$R_n(A) \le O(\sqrt{\frac{d\log n}{n}}).$$

For linear model, by doing direct calculation, we can show that $R_n(A) \asymp \sqrt{d/n}$. So the above bound is sub-optimal by a factor of $\sqrt{\log n}$. In general, this sub-optimality can go beyond log factors for nonparametric function classes.

Multi-step discretization: Potential looseness of one-step discretization: $(a - \pi(a))$ may contain some structures, but we just bound it using Cauchy–Schwarz.

Now let us consider a sequence of discrete approximations to A (not necessarily nested).

$$\{0\} = A_0, A_1, \cdots, A_m, \cdots$$

Denote $\pi_m(a) := \arg\min_{a^{(i)} \in A_m} \left\|a - a^{(i)}\right\|_n$ as the projection of a onto the m-th covering set.

For $m \ge 1$, choose A_m to be a δ_m-covering of A with respect to $\|\cdot\|_n$, where $\delta_m \downarrow 0$. At scale zero we only require $\sup_{a\in A}\|a\|_n \le \delta_0$. For each $a \in A$, we have the decomposition

$$a = \sum_{m=0}^{\infty} \big(\pi_{m+1}(a) - \pi_m(a)\big).$$

More precisely, a finite sum through scale M has remainder $a - \pi_M(a)$, whose normalized norm is at most δ_M. Cauchy–Schwarz bounds its contribution to $R_n(A)$ by δ_M, uniformly in a, so the remainder vanishes as $M \to \infty$. So we have

$$\mathbb{E}\Big[\sup_{a\in A} n^{-1}|\varepsilon^\top a|\Big] \le \sum_{m=0}^{\infty} \mathbb{E}\Big[\sup_{a\in A} n^{-1}|\varepsilon^\top(\pi_{m+1}(a) - \pi_m(a))|\Big].$$

Each term measures the complexity at scale δ_m, and each term takes supremum over a finite set. So we have

$$\mathbb{E}\Big[\sup_{a\in A} n^{-1}|\varepsilon^\top(\pi_{m+1}(a) - \pi_m(a))|\Big] \le (\delta_m + \delta_{m+1}) \cdot \sqrt{\frac{2\log\big(2|A_m|\,|A_{m+1}|\big)}{n}}.$$

By manipulating this type of bounds carefully, we can arrive at a nice form

Theorem 6.3 (Dudley's chaining bound)**.** *For a nonempty bounded set $A \subseteq \mathbb{R}^n$ containing 0, there exists a universal constant $c > 0$ such that*

$$R_n(A) \le c \int_0^{\operatorname{diam}(A)} \sqrt{\frac{\log\big(2N(A; \|\cdot\|_n, \epsilon)\big)}{n}}\, d\epsilon.$$

Proof: Let D be the diameter of A under $\|\cdot\|_n$. If $D = 0$, then $A = \{0\}$ and the claim is immediate. Otherwise, let $\delta_m = 2^{-m}D$, choose minimum-cardinality covers for $m \ge 1$, and set $A_0 = \{0\}$. The anchor ensures $\|a - \pi_0(a)\|_n \le D$. We have

$$\delta_m + \delta_{m+1} = 3 \cdot 2^{-m-1}D, \qquad 2|A_m|\,|A_{m+1}| \le \big(2N(A; \|\cdot\|_n, \delta_{m+1})\big)^2.$$

Substituting to the bound, we have

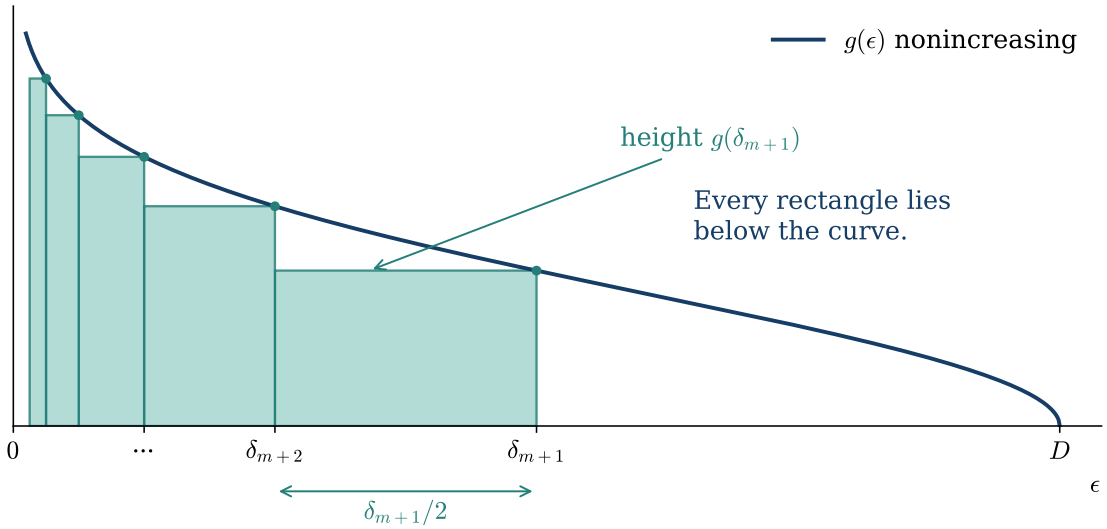
$$R_n(A) \le \sum_{m=0}^{\infty} 3 \cdot 2^{-m-1} D \cdot \sqrt{\frac{4\log\big(2N(A; \|\cdot\|_n, \delta_{m+1})\big)}{n}}.$$

To compare this dyadic sum with an integral, put

$$g(\epsilon) = \sqrt{\frac{\log\big(2N(A; \|\cdot\|_n, \epsilon)\big)}{n}}.$$

The covering number, and hence g, is nonincreasing in ϵ. On $[\delta_{m+2}, \delta_{m+1}]$, we therefore have $g(\epsilon) \ge g(\delta_{m+1})$. Since $\delta_{m+2} = \delta_{m+1}/2$, the rectangle shown below gives

$$\delta_{m+1} g(\delta_{m+1}) \le 2 \int_{\delta_{m+2}}^{\delta_{m+1}} g(\epsilon)\, d\epsilon. \tag{6.1}$$



Right-endpoint rectangles lie below a nonincreasing curve. The displayed curve is illustrative; the same area comparison applies to the entropy function g above.

The preceding chaining sum is $6\sum_{m\geq 0}\delta_{m+1}g(\delta_{m+1})$. Summing (6.1) over disjoint adjacent intervals yields

$$\begin{aligned} 6\sum_{m=0}^{\infty}\delta_{m+1}g(\delta_{m+1}) &\leq 12\int_0^{D/2} g(\epsilon)\,d\epsilon \\ &\leq 12\int_0^{D} g(\epsilon)\,d\epsilon. \end{aligned}$$

This proves the sum-to-integral comparison, including the constant factor. For an infinite sum, apply the comparison first to finite partial sums and then use monotone convergence; if the integral diverges, the inequality remains valid but is uninformative.

Remarks:

- For bounded $A \subseteq \mathbb{R}^n$, the entropy integral is finite; divergence becomes possible in general function spaces or for a worst-case uniform entropy integral. A truncated bound is also useful quantitatively: use finitely many scales and bound the residual by Cauchy–Schwarz to obtain

$$R_n(A) \leq 4\delta + c\int_{\delta}^{\operatorname{diam}(A)} \sqrt{\frac{\log\left(2N(A;\|\cdot\|_n,\epsilon)\right)}{n}}\,d\epsilon, \qquad 0<\delta\leq D.$$

  For $\delta < D/2$, stop at a scale $\delta_M \in [2\delta, 4\delta)$: the residual is at most 4δ, and the rectangle comparison only uses intervals above δ. For $\delta \geq D/2$, use $R_n(A) \leq D \leq 2\delta$. You can further optimize over δ.

- Dudley's bound is sharp for many classes, but not always. Generic chaining gives sharp bounds for Gaussian processes; a universally sharp characterization for Rademacher processes requires additional ideas.

3.3 Back to function classes

Recall that we want to bound $\mathbb{E}[\sup_{f\in\mathcal{F}} |(P_n - P)f|]$. Retain the measurability and pointwise separability conditions above, and assume a measurable finite envelope $G \geq 0$ satisfying $|f(x)| \leq G(x)$ for every f and $\mathbb{E}[G^2(X)] < \infty$.

Theorem 6.4. *Let $\mathcal{F}$ and G satisfy these conditions. There is a universal constant $c > 0$ such that, whenever the entropy integral below is finite,*

$$\mathbb{E}\Big[\sup_{f\in\mathcal{F}} |(P_n - P)f|\Big] \leq c\sqrt{\frac{\mathbb{E}[G^2(X)]}{n}} \int_0^1 \sqrt{\log\Big(2\sup_Q N(\mathcal{F}; \mathbb{L}^2(Q), \|G\|_{\mathbb{L}^2(Q)}\, u)\Big)}\, du.$$

The supremum is over finitely supported probability measures Q with $\|G\|_{\mathbb{L}^2(Q)} > 0$. If G is identically zero, the left side is zero and no entropy condition is needed.

This is chaining under a worst-case distribution Q. The factor 2 inside the logarithm also accounts for taking an absolute supremum.

Proof: Given the data, let $A = \{(f(X_1), \ldots, f(X_n)) : f \in \mathcal{F}\}$ and $S = \sqrt{P_n G^2}$. Adjoining 0 does not change the absolute supremum. Moreover,

$$\operatorname{diam}(A \cup \{0\}) \leq 2S, \qquad N(A \cup \{0\}; \|\cdot\|_n, \epsilon) \leq N(\mathcal{F}; \mathbb{L}^2(P_n), \epsilon) + 1.$$

Apply the anchored chaining bound to $A \cup \{0\}$. Since $\log(2(N+1)) \leq 2\log(2N)$ and the entropy is nonincreasing in its radius, the integral over $[0, 2S]$ is at most twice the corresponding integral over $[0, S]$. Thus, conditionally on the data,

$$\mathbb{E}_\varepsilon\Big[\sup_{f\in\mathcal{F}} \Big|\frac{1}{n}\sum_{i=1}^n \varepsilon_i f(X_i)\Big| \,\Big|\, X_1, \ldots, X_n\Big] \leq \frac{c}{\sqrt{n}} \int_0^S \sqrt{\log\Big(2N(\mathcal{F}; \mathbb{L}^2(P_n), \epsilon)\Big)}\, d\epsilon.$$

If $S = 0$, every sample value is zero and this conditional complexity is zero. If $S > 0$, substitute $\epsilon = Su$. The empirical measure P_n is one of the finitely supported measures in the theorem, so its normalized entropy is bounded by the supremum over Q.

Only after taking expectation over the data do we apply symmetrization:

$$\begin{aligned}\mathbb{E}\Big[\sup_{f\in\mathcal{F}} |(P_n - P)f|\Big] &\leq 2\mathcal{R}_n(\mathcal{F}) \\ &\leq \frac{c\,\mathbb{E}[S]}{\sqrt{n}} \int_0^1 \sqrt{\log\Big(2\sup_Q N(\mathcal{F}; \mathbb{L}^2(Q), \|G\|_{\mathbb{L}^2(Q)}\, u)\Big)}\, du.\end{aligned}$$

Finally, Jensen's inequality gives $\mathbb{E}[S] \leq \sqrt{\mathbb{E}[G^2(X)]}$, proving the result. If $\mathbb{E}[G^2(X)] = 0$, the envelope vanishes almost surely and the empirical-process supremum is zero directly.